Prosthetic Running Blade

Finite Element Analysis of Fatigue Behaviour and Natural Frequencies

Yousuf Shahabuddin

FEA DE2 - April 2026

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1 Introduction

1.1 Project Brief

This report documents the finite element analysis (FEA) of a lightweight prosthetic running blade, evaluated against natural frequency and fatigue life criteria representative of real-world running loads. Running blades are spring-like prostheses designed to store and release elastic energy during high-impact activities such as sprinting and jumping. Their performance is strongly governed by material selection, geometry, and dynamic characteristics under repeated foot-strike loading. The blade was designed in Fusion360 and is a back-mounted cheetah-style configuration, making contact with the track at the foot section. Two design iterations were evaluated across two materials: unidirectional carbon-fibre reinforced polymer (CFRP) and Aluminium Alloy 2014 T4.

1.2 Approach and Methodology

The blade geometry was modelled in CAD and imported into Ansys Mechanical 2024 R2 for analysis. The preliminary design (Design 1) was evaluated for fatigue life and fundamental frequency using CFRP. A geometric iteration (Design 2) was then introduced to increase the fundamental frequency and reduce peak stress. Design 2 was evaluated in both CFRP and Aluminium to compare material suitability. For Design 1, an individual mesh refinement study was performed to establish convergence; the converged mesh settings were then applied to Design 2 and the aluminium analysis. The discussion critically evaluates model limitations using FEA theory.



(a) Running Blade V1 (b) Running Blade V2

Figure 1: Comparison of the two running blade designs

1.3 Design Criteria

The running blade designs were evaluated against the following criteria:

- **Lightweight:** < 600 g, consistent with commercially available prosthetic running blades [1].

- **Fatigue resistance:** $>$ 1 million loading cycles, equivalent to approximately 2,000 km of running distance.
- **Fundamental frequency:** $>$ 50 Hz, to avoid resonance from excitation of natural frequencies after track impact [4].

2 Methods

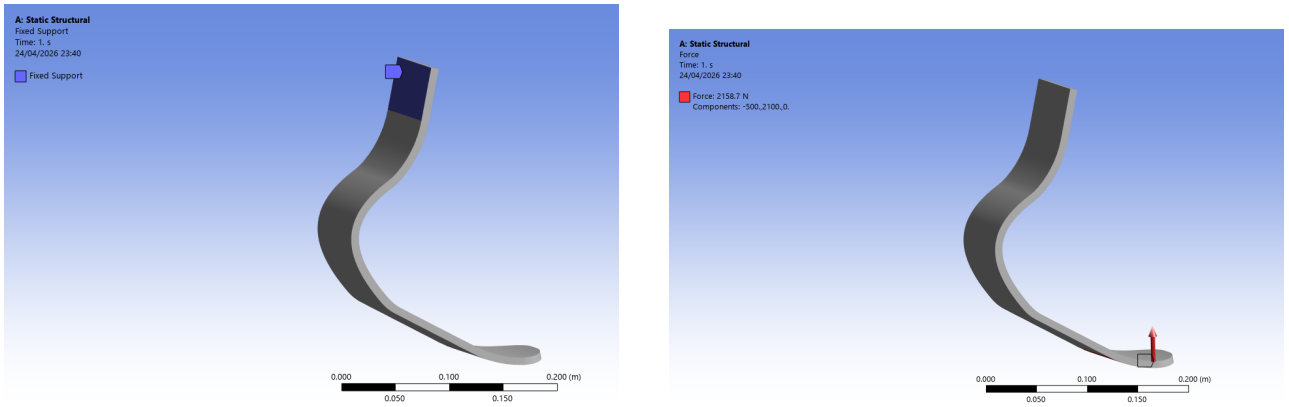
2.1 Boundary Conditions and Loads

The blade is fixed to the prosthetic socket at its upper face. In the FE model, this was represented by a Fixed Support boundary condition applied to the top face of the blade, constraining all six degrees of freedom at that surface. This assumes the socket attachment behaves as a rigid connection, consistent with the brief.

Two forces were applied simultaneously to the foot contact area at the base of the blade, representing a single foot-strike event:

- **Normal force:** 2,100 N applied vertically (positive y-direction), representing approximately three times a typical adult body weight, as specified in the project brief.
- **Friction force:** 500 N applied horizontally (negative x-direction), opposing the forward running direction, as specified in the brief.

Standard Earth Gravity (9.81 m/s^2, negative y-direction) was included in the static structural analysis to account for blade self-weight. No forces were applied during modal analyses, as these represent undamped free vibration problems. The boundary conditions are illustrated in Figure **??**.



(a) Running Blade V1 (b) Running Blade V2

Figure 2: Comparison of the two running blade designs

2.2 Material Properties

Unidirectional CFRP (Standard CF UD) properties were taken from the ACP Composites mechanical properties data sheet [2] and defined in Ansys. This was selected as it is the

industry-standard material for prosthetic running blades, with fibres oriented along the blade length to maximise flexural stiffness. An isotropic material approximation was used in the FE model, which is acknowledged as a simplification discussed further in Section 4.

Aluminium 2014 T4 properties were included within the Ansys engineering data itself, and so was used directly, as it also had the S-N curve data available.

2.3 Fatigue Study Configuration

CFRP: Standard S-N curve data is not readily available for CFRP due to the complexity of composite fatigue failure mechanisms, which involve progressive delamination, fibre breakage, and matrix cracking that are difficult to characterise with a single stress-life curve [2]. Therefore, a 3x safety factor was applied:

$$\text{UTS} \geq 3 \times \sigma_{peak} \tag{1}$$

Aluminium 2014 T4: A full stress-life fatigue analysis was performed using the Ansys Fatigue Tool. The S-N curve is typically obtained under fully reversed loading conditions ($R = -1$), where the mean stress is zero and the stress alternates symmetrically about zero. In practice, a non-zero mean stress significantly affects fatigue life; a positive (tensile) mean stress accelerates crack initiation and growth, reducing fatigue life relative to the fully reversed case [4]. Since the foot-strike loading in this study is zero-based ($R = 0$), a positive mean stress equal to half the peak stress is present in every loading cycle. The S-N curve data therefore cannot be applied directly without correction.

Mean stress correction methods modify the alternating stress to an equivalent corrected value S_{ca} that accounts for the damaging effect of the mean stress, allowing comparison against the standard $R = -1$ S-N curve. The Goodman correction was selected, as the metal exhibits relatively brittle fatigue behaviour with high mean stress sensitivity, for which Goodman provides a more appropriate correction than the Gerber method . [4]. The Goodman formula is:

$$S_{ca} = \frac{S}{1 - \dfrac{\sigma_{mean}}{S_u}} \tag{2}$$

where S is the alternating stress amplitude, σ_{mean} is the mean stress, and S_u is the ultimate tensile strength. A higher mean stress produces a larger S_{ca}, predicting shorter fatigue life, consistent with physical expectations. The von Mises criterion was used as the equivalent stress measure and the design life was set to 1,000,000 cycles.

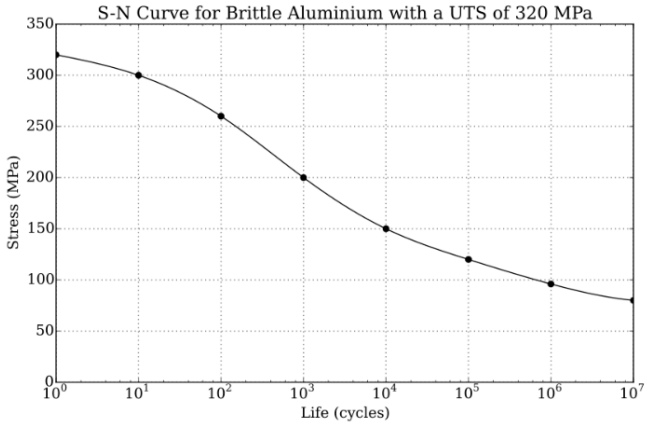


Figure 3: Aluminium 2014 T4 S-N Curve

2.4 Frequency Study Configuration

Modal analyses were performed to extract the natural frequencies and mode shapes of each blade configuration. The same geometry, material, fixed support, and mesh were used as in the corresponding static structural analysis. The first two natural frequencies and corresponding mode shapes were extracted for each configuration. Some deformation is required to return the energy to the user during each stride, but should be kept at a minimum.

3 Results

3.1 Sanity Checks

Three sanity checks were performed prior to analysis to verify model correctness.

Reaction force check: A Force Reaction probe was applied at the Fixed Support. The resultant reaction force was 2,158.7 N, composed of 500 N in the x-direction, $-2{,}100$ N in the y-direction, and 4.51×10^{-6} N in the z-direction. The analytical resultant of the applied loads is:

$$F_{resultant} = \sqrt{500^2 + 2100^2} = 2{,}158.1 \text{ N} \tag{3}$$

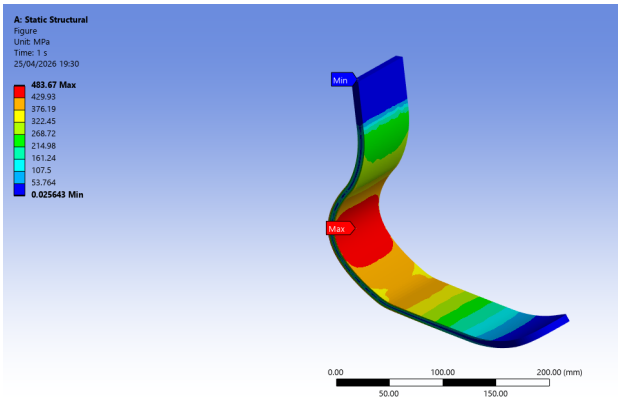
The discrepancy between the FEA reaction force and the analytical value is 0.03%, confirming correct application of boundary conditions and loading.

Deformation direction check: The total deformation plot was used to verify physically realistic blade behaviour. The foot area displaced downward and in the direction of friction, consistent with expectations. The fixed support showed zero displacement as required.

Stress location check: The peak von Mises stress was located at the inner radius of the blade curve, consistent with classical curved beam bending theory, where the highest bending stress occurs at the point of greatest curvature [4].

Details of "Force Reaction"	
Location Method	Boundary Condition
Boundary Condition	Fixed Support
Orientation	Global Coordinate System
Suppressed	No
Options	
Result Selection	All
Display Time	End Time
Results	
Maximum Value Over Time	
X Axis	500. N
Y Axis	-2100. N
Z Axis	-2.4824e-005 N
Total	2158.7 N
Minimum Value Over Time	

(a) Force Reaction probe result.



(b) Total deformation plot showing realistic blade deflection.

Figure 4: Sanity check results confirming correct model setup.

3.2 Design 1 - CFRP

3.2.1 Mesh Refinement Study

Quadratic elements were used throughout all analyses. They accurately represent curved geometries and capture steep stress gradients through quadratic interpolation of the displacement field [4]. A global mesh refinement study was conducted using five element sizes. For each mesh, the peak von Mises stress was recorded and the convergence error calculated:

$$\varepsilon = \left| \frac{\sigma_n - \sigma_{n-1}}{\sigma_{n-1}} \right| \times 100\% \tag{4}$$

Table 1: Global mesh refinement study - Design 1 CFRP.

Global Element Size (mm)	Peak von Mises Stress (MPa)	Convergence Error (%)
15	525.93	-
10	473.25	10.02
7	490.01	3.54
5	487.74	0.46
2	484.21	0.72

A global element size of 5 mm was selected, as the convergence error between 5 mm and 2 mm was 0.72%, within the 1% target threshold. A 2 mm global mesh was not used for subsequent analyses due to element count restrictions imposed by the Ansys Student licence, however it would not likely have been necessary anyway.

Following global convergence, a local mesh refinement study was conducted at the identified stress concentration (inner curve radius), where the steep stress gradient requires finer discretisation to accurately resolve the peak stress field:

Table 2: Local mesh refinement study at inner curve radius - Design 1 CFRP.

Local Element Size (mm)	Peak von-Mises Stress (MPa)	Convergence Error (%)
5 (global only)	487.74	-
4	403.30	5.09
3	383.81	0.09

A local element size of 3 mm was selected as the final mesh configuration. The convergence error of 0.09% falls within the 1% target; further refinement globally was not feasible, again within the Ansys Student licence, however still produces $< 1\%$ error. This represents a small source of discretisation error, discussed further in Section 4. The same converged mesh settings (5 mm global, 3 mm local at stress concentration) were applied to Design 2 and the aluminium analysis to ensure comparability of results.

Table 3: Mesh quality metrics - Design 1 CFRP (5 mm global, 3 mm local refinement).

Metric	Minimum	Maximum	Average	Std Dev	Assessment
Aspect Ratio (simulated)	1.1893	11.977	2.4074	0.7954	See note (a)
Aspect Ratio (recommended)	1 (ideal)	<10	-	-	-
Jacobian Ratio (simulated)	0.7555	1.0000	0.9849	0.0225	See note (b)
Jacobian Ratio (ideal)	1.0 (perfect)		-	-	-

(a) Average aspect ratio of 2.41 indicates the majority of elements are well-proportioned. The maximum value of 11.98 marginally exceeds the recommended threshold of 10, attributable to a small number of elements at thin curved boundaries where the mesher cannot place ideal tetrahedra. These elements are not located in the critical stress concentration region.
(b) Average Jacobian ratio of 0.985 confirms minimal distortion of quadratic element mid-side nodes throughout the mesh, indicating accurate geometric representation of the curved blade surface [4].

Following convergence, mesh quality was evaluated to determine discretisation accuracy. The aspect ratio exhibited a minimum of 1.19, an average of 2.41, a maximum of 11.98, and a standard deviation of 0.80. The average aspect ratio reflects acceptable element quality, although the maximum value slightly exceeds the recommended threshold of 10. This result indicates the presence of a small number of distorted elements near thin, curved blade boundaries. These elements constitute a minor portion of the mesh and are not situated in regions of critical stress.

The Jacobian ratio ranged from 0.755 to 1.0, with an average of 0.985 and a standard deviation of 0.022. Because values near 1 indicate low distortion, the quadratic tetrahedral elements accurately represent the curved blade geometry, exhibiting minimal mid-side node distortion. [4]

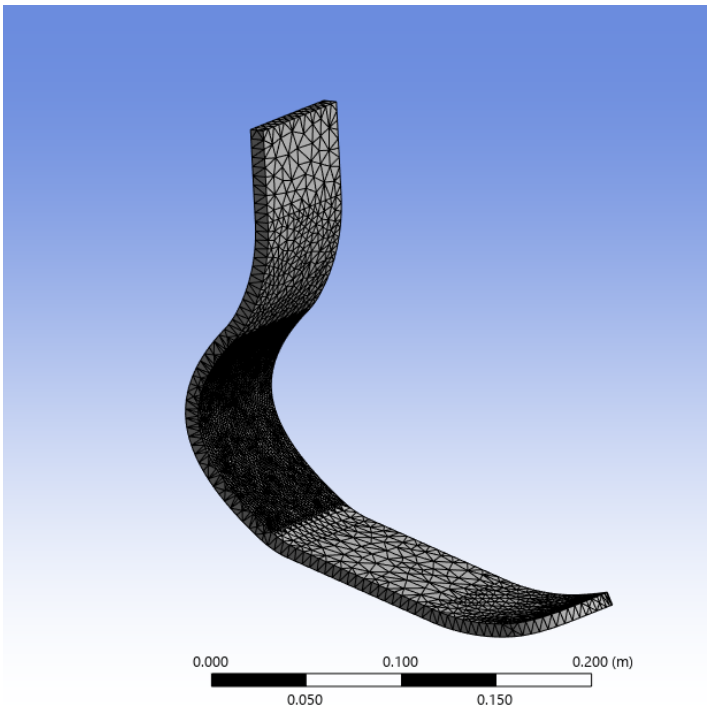


Figure 5: Final converged mesh for Design 1

3.2.2 Fatigue Study Results - CFRP Design 1

The peak von Mises stress from the static structural analysis with the converged mesh was 403.30 MPa. The safety factor criterion requires:

$$\text{UTS} \geq 3 \times \sigma_{peak} \tag{5}$$

$$3 \times 403.30 = 1{,}209.9 \text{ MPa} \tag{6}$$

Since UTS = 1,500 MPa > 1,209.9 MPa, **Design 1 in CFRP satisfies the fatigue criterion** and is predicted to withstand in excess of 1 million loading cycles. The safety margin is:

$$\text{Safety margin} = \frac{1{,}500}{403.30} = 3.72 \tag{7}$$

This meets the requirement. The maximum total deformation under loading was 80.0 mm, consistent with the spring-like deflection behaviour of real prosthetic running blades, however optimal running blades typically deform by $\tilde{3}$0mm, and so this may be cause for concern [5].

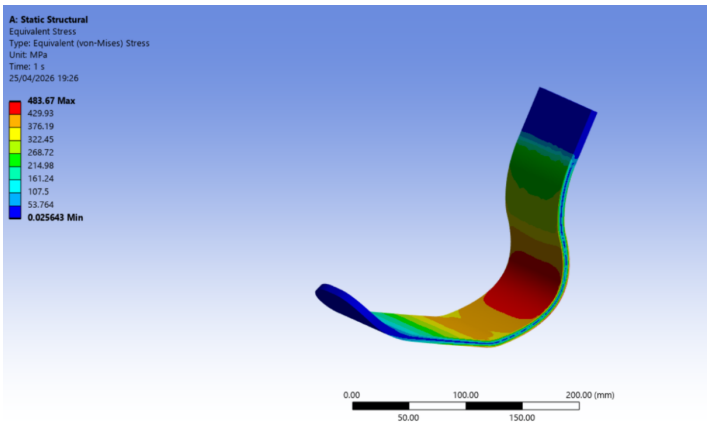


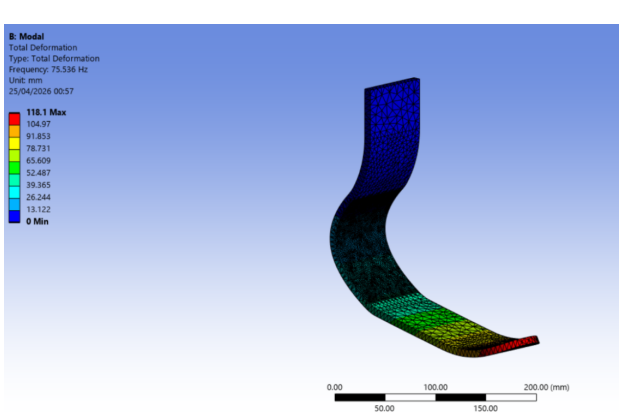
Figure 6: Von Mises stress distribution for Design 1 CFRP. Peak stress 403.30 MPa located at inner curve radius.

3.2.3 Frequency Study Results - CFRP Design 1

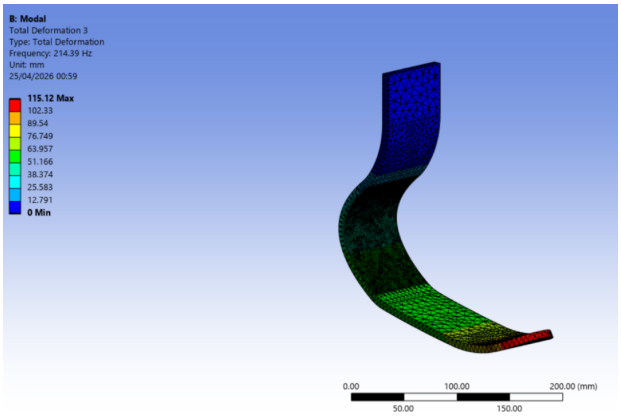
Table 4: Natural frequencies for Design 1 - CFRP.

Mode	Frequency (Hz)	Mode Description
1	75.5	Fundamental lateral bending
3	134.0	Second bending / torsional

The fundamental frequency of 75.5 Hz **exceeds the 50 Hz criteria**. Mode 1 represents lateral bending of the blade tip relative to the fixed support. Mode 3 represents out-of-plane bending, in which the blade deflects laterally. From vibration theory, higher stiffness k in a given direction produces a higher natural frequency $\omega = \sqrt{k/m}$ for that mode [4].



(a) Mode 1 - 75.5 Hz.



(b) Mode 2 - 134.0 Hz.

Figure 7: Mode shapes for Design 1 CFRP.

3.3 Design Iteration Rationale

Although Design 1 met both the fatigue and frequency criteria, ot was iterated improve fundamental frequency and reduce peak stress. Design 1 CFRP weighed 371.1g, and as this was

well within the criteria defined, I opted to increase the thickness of the blade as there was headroom with the weight.

Design 2 CFRP: 422.4g Design 2 Aluminium: 859.2g

The following modifications were made:

- Increased curve radius at the main bend, reducing stress concentration at the inner surface.
- Variable thickness profile with an overall increase in blade thickness.
- Slight lengthening of the tip and shortening of the mid-section to redistribute stiffness.

From vibration theory, $\omega \propto \sqrt{k/m}$ [4]. Increasing blade thickness raises the second moment of area I, directly increasing flexural stiffness $k \propto EI/L^3$. Since stiffness scales with the cube of thickness ($I \propto t^3$) while mass scales only linearly ($m \propto t$), the net effect is an increase in natural frequency. Simultaneously, increasing the curve radius reduces the geometric stress concentration factor, lowering peak bending stress.

3.4 Design 2 - CFRP

3.4.1 Fatigue Study Results - CFRP Design 2

The peak von Mises stress for Design 2 in CFRP was 302.02 MPa. Applying a 3x safety factor:

$$3 \times 302.02 = 906.1 \text{ MPa} \tag{8}$$

Since UTS = 1,500 MPa > 906.1 MPa, **Design 2 in CFRP satisfies the fatigue criterion**. The safety margin is:

$$\text{Safety margin} = \frac{1{,}500}{302.02} = 4.97 \tag{9}$$

This represents a substantial improvement over Design 1 (safety margin 3.72). The peak stress reduced by 25.1% relative to Design 1, attributable to the increased curve radius reducing stress concentration and the variable increased thickness. Maximum total deformation was 102.7 mm, slightly higher than Design 1 due to the lengthened tip increasing the effective moment arm, and to keep it more in proportion to running blades.

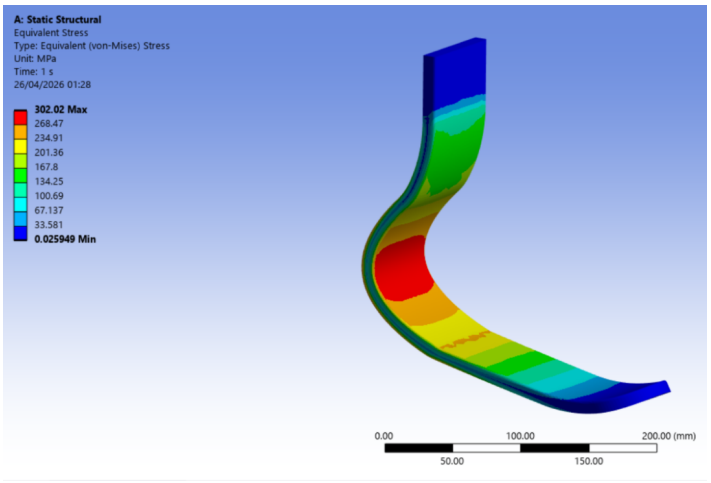


Figure 8: Von Mises stress distribution for Design 2 CFRP. Peak stress 302.02 MPa, a 25.1% reduction from Design 1.

3.4.2 Frequency Study Results - CFRP Design 2

Table 5: Natural frequencies for Design 2 - CFRP.

Mode	Frequency (Hz)	Mode Description
1	97.4	Fundamental lateral bending
2	170.4	Second bending / torsional

The fundamental frequency increased from 75.5 Hz to 97.4 Hz, a 29.0% improvement. This is consistent with the theoretical prediction: increasing thickness raises $I \propto t^3$, increasing stiffness k by a larger factor than mass $m \propto t$, giving a net increase in $\omega = \sqrt{k/m}$. Design 2 comfortably exceeds the 50 Hz criteria.

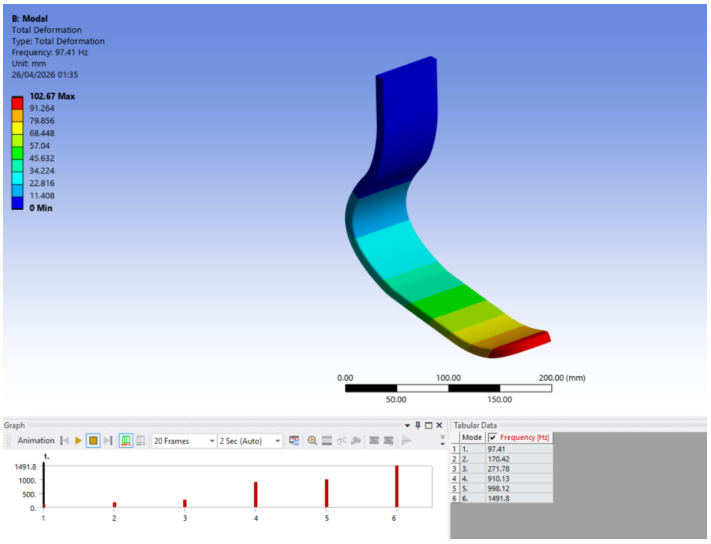


Figure 9: Mode shape 1, natural frequency

3.5 Design 2 - Aluminium Alloy 2014 T4

3.5.1 Fatigue Study Results - Aluminium Design 2

The peak von Mises stress for the aluminium blade was 314.01 MPa. The yield strength of Aluminium 2014 T4 is 270 MPa [3]. Since the peak stress (314.01 MPa) exceeds the yield strength, local plastic deformation occurs at the critical location. The stress-life fatigue approach assumes linear elastic behaviour; when this is transgressed, the S-N results in the plastic zone must be interpreted with caution as the approach is no longer strictly valid.

A manual Goodman mean stress correction was performed as a pre-simulation check. For zero-based loading:

$$\sigma_{mean} = S = \frac{\sigma_{peak}}{2} = \frac{314.01}{2} = 157.0 \text{ MPa} \tag{10}$$

$$S_{ca} = \frac{S}{1 - \frac{\sigma_{mean}}{S_u}} = \frac{157.0}{1 - \frac{157.0}{430}} = \frac{157.0}{0.635} = 247.2 \text{ MPa} \tag{11}$$

Comparing the corrected alternating stress $S_{ca} = 247.2$ MPa against the high-cycle fatigue strength of Aluminium 2014 T4 of 140 MPa [3]: since 247.2 MPa $>$ 140 MPa, the Goodman correction predicts fatigue failure, consistent with the Ansys fatigue tool results.

Table 6: Goodman mean stress correction – Aluminium 2014 T4 Design 2.

Material	σ_{peak} **(MPa)**	**UTS (MPa)**	σ_{mean} **(MPa)**	S **(MPa)**	S_{ca} **(MPa)**	**Fatigue Strength (MPa)**	**Pass?**
Al 2014 T4	314.01	430	157.0	157.0	247.2	140	**No**

The Ansys Fatigue Tool, using Goodman correction with zero-based loading and the Al 2014 T4 S-N curve, reported a minimum fatigue life of 26,670 cycles in the localised region of peak stress, with an average life across the blade of approximately 73 million cycles (although irrelevant seeing as my minimum is 26,670 cycles The equivalent alternating stress reported by Ansys was 211.17 MPa. The minimum life of 26,670 cycles does not satisfy the 1 million cycle fatigue criteria. Maximum total deformation under loading was 78.2 mm.

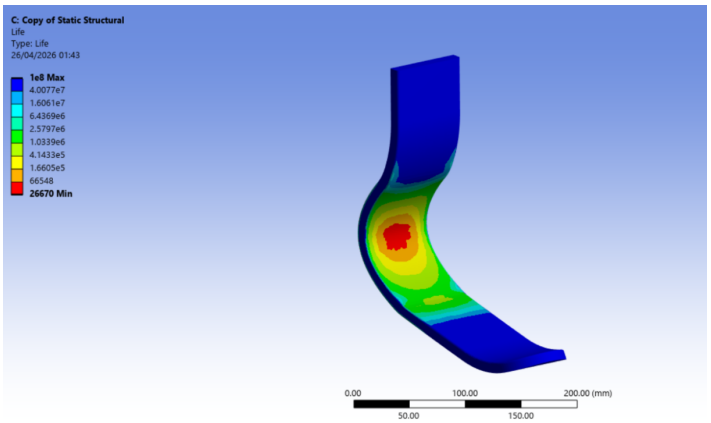


Figure 10: Fatigue life contour for Design 2 Aluminium 2014 T4. Minimum life 26,670 cycles at inner curve radius. Red zone indicates region of earliest fatigue failure.

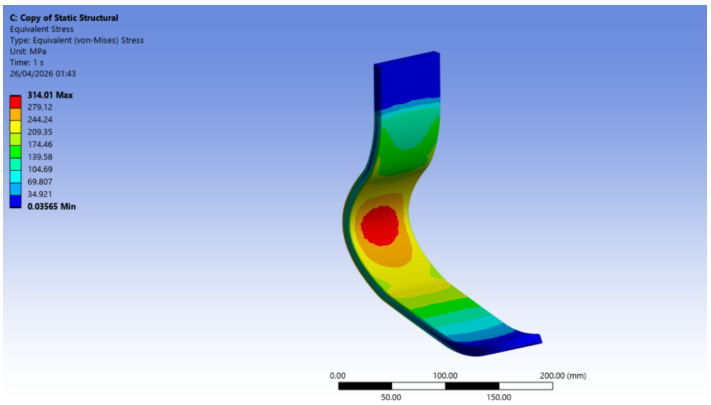


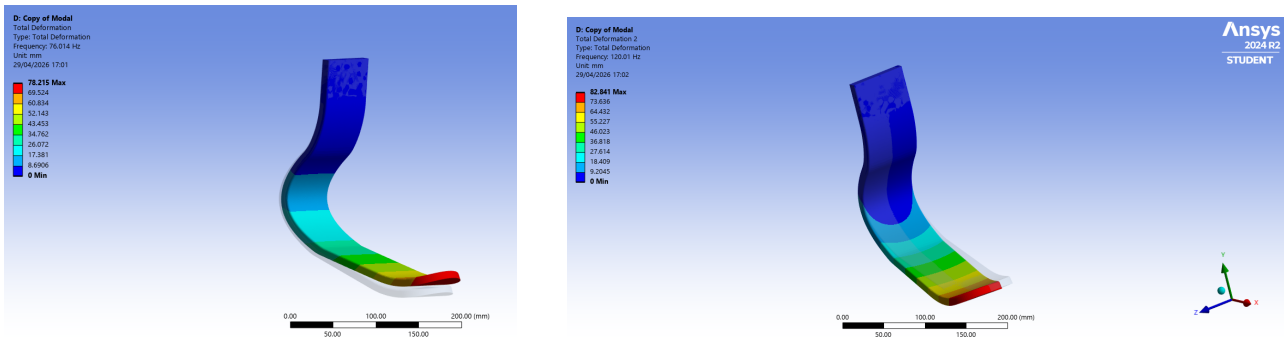
Figure 11: Von Mises stress distribution for Design 2 Aluminium 2014 T4. Peak stress 314.01 MPa exceeds yield strength of 270 MPa.

3.5.2 Frequency Study Results - Aluminium Design 2

Table 7: Natural frequencies for Design 2 - Aluminium Alloy 2014 T4.

Mode	Frequency (Hz)	Mode Description
1	76.0	Fundamental lateral bending
2	120.0	Second bending / torsional

The fundamental frequency of the aluminium blade (76.0 Hz) is lower than the equivalent CFRP blade (97.4 Hz). Both materials have similar Young's moduli, however the density of aluminium is significantly higher than CFRP, increasing mass m for the same geometry and therefore reducing $\omega = \sqrt{k/m}$. The difference in Young's modulus partially offsets this effect, as the higher stiffness of CFRP (135 GPa vs 72 GPa) further increases k relative to aluminium. The aluminium blade **exceeds the 50 Hz frequency criterion** at 76.0 Hz.



(a) Mode 1 - 76.0 Hz.

(b) Mode 2 - 120.0 Hz.

Figure 12: Mode shapes for Design 2 Aluminium 2014 T4.

4 Discussion

4.1 Summary of Results

Table 8: Summary of all results across design configurations and materials.

Design	**Peak Stress (MPa)**	**Strength/Fatigue Check**	**Freq. (Hz)**	**Pass?**
CFRP V1	403.30	Pass: $1500 \geq 3 \times 403.30$	75.5	Yes
CFRP V2	302.02	Pass: $1500 \geq 3 \times 302.02$	97.4	Yes
Aluminium	314.01	Fail: min. life 26,670 cycles	76.0	No

Both CFRP configurations satisfied all design criteria. The geometric iteration from Design 1 to Design 2 reduced peak stress by 25.1% and increased fundamental frequency by 29.0%, while also increasing the safety margin from 3.72 to 4.97. The aluminium configuration failed, with peak stresses exceeding the material yield strength and a minimum predicted life of only 26,670 cycles. It also failed with regards to weight, weighing 259.22g over the limit. This confirms that CFRP is the more appropriate material for prosthetic running blades, consistent with its universal adoption in elite prosthetic blade manufacturing due to its superior specific strength and excellent fatigue resistance under cyclic loading.

4.2 Model Accuracy and Limitations

The module handbook identifies three principal sources of error in FEA: modelling error, discretisation error, and numerical error [4].

Modelling error arises from idealisations in representing the physical problem. The most significant modelling assumption is the use of an isotropic material model for CFRP. Real unidirectional CFRP is strongly orthotropic. The isotropic approximation used here cannot entirely capture this directional dependence, however it has been modelled to be aligned as well as possible.

The Fixed Support boundary condition assumes a perfectly rigid socket attachment. In reality, the socket interface is compliant and energy-absorbing, which would reduce stress concentrations at the attachment region. Additionally, the applied loads are static approximations of a dynamic impact event. The foot-strike force varies with time, athlete mass, running speed, and surface stiffness. Many blades also have rubber attachments, which would improve shock absorption. A static analysis ignores inertial and damping effects, and dynamic amplification factors could increase the effective alternating stress beyond the static values used here. This is a further source of modelling error particularly relevant to the fatigue assessment.

Discretisation error arises from the finite element approximation of the mathematical model. The local mesh convergence error of 0.46 % and 0.09% at Design 1 indicates that the peak stress results carry a minute discretisation uncertainty. It could have been improved further however due tp element count restrictions imposed by the Ansys Student licence, this was not possible.

The use of quadratic tetrahedral elements mitigates discretisation error relative to linear elements, as they more accurately capture curved geometry and higher-order displacement fields [4]. The fixed mesh settings applied to Design 2 and the aluminium analysis introduce a further assumption that the converged mesh from Design 1 remains appropriate for the modified geometries; ideally, a separate convergence study would be performed for each configuration. However, changes made were not major, such as adding ribs or other materials, as this would require a much deeper understanding of the mechanics of prostheses, which was not within the scope of this project (although some research has been carried out).

Mesh quality metrics confirmed that the majority of elements had acceptable aspect ratios (average 2.41) and excellent Jacobian ratios (average 0.985), though a small number of high aspect ratio elements (maximum 11.98) were present at thin curved boundaries.

Numerical error arises from finite-precision arithmetic in the solver. For the stress magnitudes involved (hundreds of MPa), numerical precision errors are of the order of 0.01 MPa, which is negligible relative to the discretisation and modelling errors identified above.

Despite these limitations, the FE models produce physically consistent and internally coherent results that support the conclusion that CFRP Design 2 is the most suitable configuration. Validation against physical testing would be required before these models could be used for design certification purposes.

5 References

References

[1] Ottobock (2026). *1E91 Taleo Running Blade.* [online] Available at: https://www.ottobock.com/en-gb/product/1E91-60984 [Accessed 30 April 2026].

[2] ACP Composites (2023). *Mechanical Properties of Carbon Fiber Composite Materials.* [online] Available at: https://acpcomposites.com/wp-content/uploads/2023/12/Mechanical-Properties-of-Carbon-Fiber-Composite-Materials.pdf [Accessed 30 April 2026].

[3] MakeItFrom.com (2020). *2014-T4 Aluminum.* [online] Available at: https://www.makeitfrom.com/material-properties/2014-T4-Aluminum [Accessed 30 April 2026].

[4] Ghajari, M. (2026). *Finite Element Analysis Module Handbook and Lecture Notes v1.6.* Dyson School of Design Engineering, Imperial College London.

[5] https://ein.org.pl/pdf-172668-94620?filename=Assessment-of-a-Carbon-Fi.pdf

6 Appendices

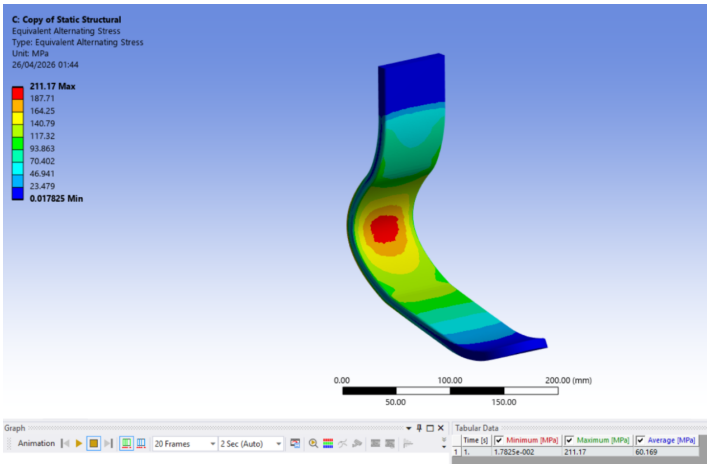


Figure 13: Aluminium, alternating stress

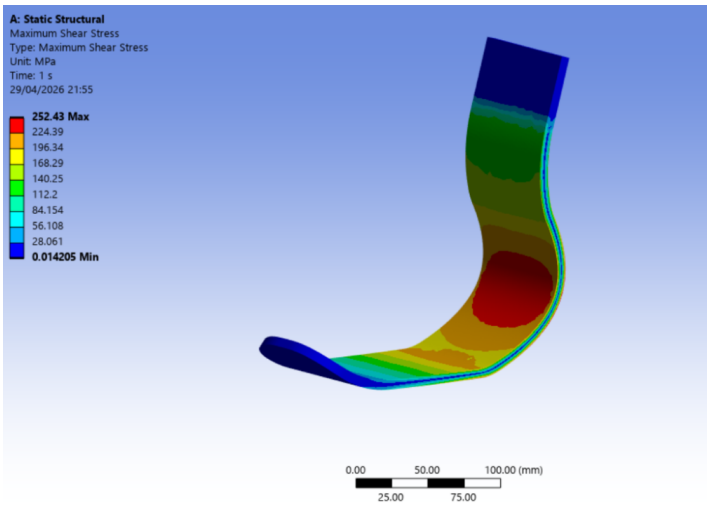


Figure 14: Design 1 CFRP Shear Stress

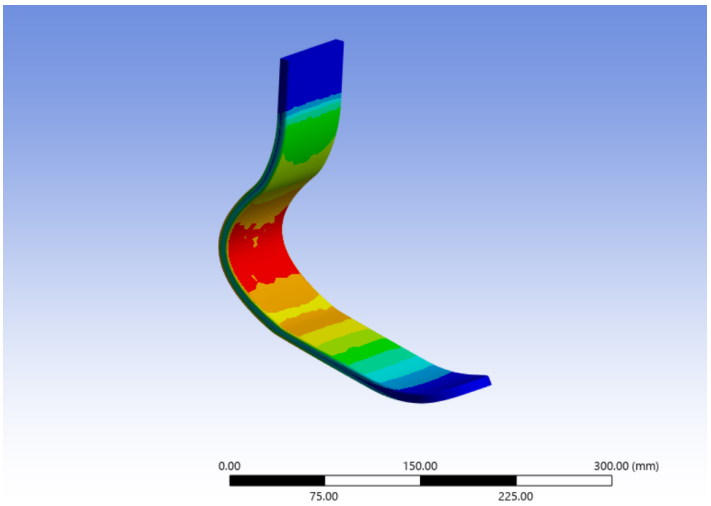


Figure 15: Design 2 CFRP pre-mesh refinement

AI Usage Statement: I used Claude to assist with structuring and editing sections of this report to keep it within the word limit. I also used ChatGPT to convert maths equations into TeX for better formatting.